

REQUIREMENTS SPECIFICATION

rqtk — Requirements Toolkit

Self-hosted requirements for the rqtk CLI and core
library

VERSION

0.1.0

DATE

May 11, 2026

CLASSIFICATION

Open Source

MISSION PHASE

Phase-A

ORGANIZATION

rqtk

REQUIREMENTS

24 total

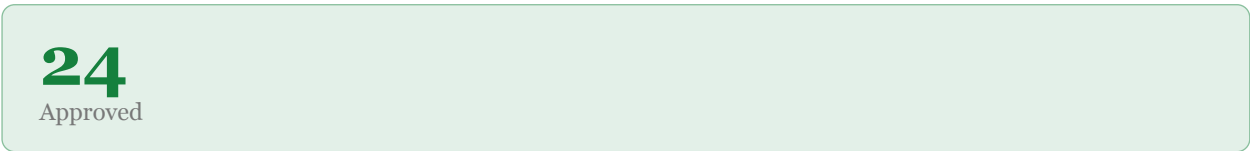
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1 Executive Summary

1.1 Status Overview



1.2 Coverage by Category

Category	Description	Total	Approved	Draft
CLI	Requirements for the rqtk CLI binary	12	12	0
CORE	Requirements for the rqtk-core library crate	4	4	0
SYS	Top-level product requirements derived from user and stakeholder needs	8	8	0

2 SYS — System

Top-level product requirements derived from user and stakeholder needs

RQTK-SYS-0001 **Requirement Lifecycle Management** Approved Critical Functional

STATEMENT

rqt shall support the full lifecycle of individual requirements, including creation, update, validation, versioning, and deprecation.

RATIONALE

A requirement that cannot be tracked from creation to deprecation loses traceability and becomes a maintenance liability; lifecycle coverage is the minimum unit of value for any requirements tool.

CATEGORY SYS	HASH 98fcc22e	CRITICALITY Mission-Critical	VERIFY Test
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VERIFICATION ACTIVITIES

VA-SYS-001-01 Lifecycle round-trip test Planned

Expected: Field-for-field equality after round-trip for each state value.

VA-SYS-001-02 Requirement update verification Planned

Expected: State field equals 'Review'; all other fields retain their initial values.

RQTK-SYS-0002 **TOML-Based File Storage** Approved Critical Design

STATEMENT

rqt shall store each requirement as a single standalone TOML file conforming to the TOML 1.0 specification.

RATIONALE

One-file-per-requirement isolates merge conflicts to individual requirements; TOML 1.0 has stable semantics that survive editor reformatting and are parseable without preprocessing.

CATEGORY SYS	HASH df906ed4	CRITICALITY Mission-Critical	VERIFY Inspection
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VERIFICATION ACTIVITIES

VA-SYS-002-01 File format inspection

Planned

Expected: File is valid TOML 1.0, contains exactly one [requirement] table, and is schema-conformant.

RQTK-SYS-0003 Requirement Traceability Graph

Approved

Critical

Functional

STATEMENT

rqt shall maintain a directed acyclic graph of parent–child relationships between requirements.

RATIONALE

A formal graph model is the prerequisite for automated traceability traversal, coverage analysis, and graph export; without it, requirement relationships exist only as informal annotations.

CATEGORY
SYS**HASH**
291f3f95**CRITICALITY**
Mission-Critical**VERIFY**
Test

VERIFICATION ACTIVITIES

VA-SYS-003-01 Graph round-trip test

Planned

Expected: Edge count equals the total number of parent references across all loaded requirements; no edges are missing or duplicated.

RQTK-SYS-0004 Policy-Driven Validation

Approved

Critical

Functional

STATEMENT

rqt shall evaluate all loaded requirements against the lint rules defined in the project configuration.

RATIONALE

Config-driven evaluation is the core value proposition; it decouples domain vocabulary (allowed states, categories, types) from the tool and lets any project enforce its own policy without modifying rqt.

CATEGORY
SYS**HASH**
b9c453f7**CRITICALITY**
Mission-Critical**VERIFY**
Test

VERIFICATION ACTIVITIES

VA-SYS-004-01 Lint rule coverage matrix

Planned

Expected: All 18 codes reachable; no false positives on the clean firesat-obc fixture.

RQTK-SYS-0005 Multi-Format Export

Approved

High

Functional

STATEMENT

rqt shall export the validated requirement set in JSON, CSV, and Markdown formats, writing output either to stdout or to a user-specified file path.

RATIONALE

Downstream tools (spreadsheets, document generators, CI dashboards) consume different formats; supporting all three removes the need for bespoke conversion scripts and keeps rqt as the single source of truth. Exporting only the validated set prevents policy-violating data from propagating into downstream artefacts.

CATEGORY
SYSHASH
20c0561fCRITICALITY
Non-CriticalVERIFY
Test

VERIFICATION ACTIVITIES

VA-SYS-005-01 Export format validation

Planned

Expected: All three formats valid and contain the correct number of requirement entries.

RQTK-SYS-0006 VCS Merge Compatibility

Approved

High

Design

STATEMENT

Each rqt requirement file shall contain only plain-text UTF-8 content such that a merge conflict between concurrent edits to different fields of the same file is resolvable by standard three-way VCS merge tooling without rqt.

RATIONALE

Plain-text, field-per-line TOML means git's three-way merge can resolve most concurrent edits automatically; binary or encoded content would make every conflict a manual, tool-assisted exercise.

CATEGORY
SYSHASH
7eabd787CRITICALITY
Non-CriticalVERIFY
Demonstration

VERIFICATION ACTIVITIES

VA-SYS-006-01 Concurrent field edit merge demonstration

Planned

Expected: No conflict markers; rqt lint exits 0 on the merged file.

RQTK-SYS-0007 Traceability Graph Defect Detection

Approved

Critical

Functional

STATEMENT

rqt shall detect and report, at validation time, all cycles, parentless non-root requirements (orphans), and broken parent references within the traceability graph.

RATIONALE

Undetected cycles or dangling parent links silently invalidate traceability matrices; automated defect detection prevents those errors from reaching reviews or audits.

CATEGORY	HASH	CRITICALITY	VERIFY
SYS	1f8ccfa4	Mission-Critical	Test

VERIFICATION ACTIVITIES

VA-SYS-007-01 Cycle detection test

Planned

Expected: Exit code 2; output contains RQ017.

VA-SYS-007-02 Broken reference test

Planned

Expected: Exit code 2; output contains RQ015.

VA-SYS-007-03 Orphan detection test

Planned

Expected: Exit code 2; orphan requirement ID present in output.

RQTK-SYS-0008 Structured Violation Reporting

Approved

Critical

Functional

STATEMENT

rqt shall report each lint violation as a record containing a stable rule code, a severity level, the affected requirement ID, and the source file path.

RATIONALE

Stable rule codes enable CI gates, suppression lists, and audit trails; structured output allows downstream tooling to parse and filter violations without fragile string matching.

CATEGORY	HASH	CRITICALITY	VERIFY
SYS	3f01b706	Mission-Critical	Test

VERIFICATION ACTIVITIES

VA-SYS-008-01 Violation record structure test

Planned

Expected: Output record contains rule code 'RQ007', severity 'Error', the fixture requirement ID, and the file path of the offending TOML file.

3 CLI — Command-Line Interface

Requirements for the rqt CLI binary

RQTK-CLI-0001 Add Requirement

ApprovedHighFunctional

STATEMENT

The rqt add command shall generate a requirement TOML file with an auto-incremented ID, with default field values for state, priority, verification method, level, and phase drawn from the project configuration.

RATIONALE

Manual ID assignment is error-prone and slows authoring; auto-increment with config defaults ensures every new requirement starts in a valid, policy-conformant state without requiring the engineer to memorise allowed values.

CATEGORY
CLI

HASH
b17a4a48

CRITICALITY
Non-Critical

VERIFY
Test

PARENTS

RQTK-SYS-0001 · RQTK-SYS-0002

VERIFICATION ACTIVITIES

VA-CLI-001-01 ID auto-increment test

Planned

Expected: File created with ID RQTK-SYS-0003.

RQTK-CLI-0002 Lint Command Output

ApprovedCriticalFunctional

STATEMENT

The rqt lint command shall evaluate all requirements in the configured requirements directory against the policy rules defined in rqt.toml and print each violation as a single line containing the rule code, severity, requirement ID, and file path.

RATIONALE

Structured, one-violation-per-line output allows CI pipelines and downstream tools to parse results without fragile string matching; using the configured requirements directory keeps behavior reproducible across environments.

CATEGORY CLI	HASH 68645d1b	CRITICALITY Non-Critical	VERIFY Test
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PARENTS
[RQTK-SYS-0004](#) · [RQTK-SYS-0008](#)

VERIFICATION ACTIVITIES

VA-CLI-002-01 Violation output format test

Planned

Expected: Violation line contains RQ007, severity, and requirement ID; clean fixture produces no output.

[RQTK-CLI-0003](#) Traceability Query

Approved

High

Functional

STATEMENT

The rqt trace command shall, given a requirement ID, print the full upward path (ancestors) and downward path (descendants) of that requirement within the traceability graph.

RATIONALE

Engineers need to answer two questions: 'what stakeholder need justifies this requirement?' (upward) and 'what lower-level requirements implement this?' (downward); a single command that shows both eliminates manual graph traversal.

CATEGORY CLI	HASH b644da2f	CRITICALITY Non-Critical	VERIFY Test
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PARENTS
[RQTK-SYS-0003](#)

VERIFICATION ACTIVITIES

VA-CLI-003-01 Bidirectional trace test

Planned

Expected: FOBC-SYS-0001 present in upward path section.

[RQTK-CLI-0004](#) Coverage Gap Detection

Approved

High

Functional

STATEMENT

The rqt coverage command shall identify and report all requirements that either have an empty `success_criteria` field or have zero verification activities defined.

RATIONALE

Verification coverage is a contractual obligation on most engineering programmes; an automated gap report prevents unverified requirements from reaching design reviews undetected.

CATEGORY CLI	HASH 66bcf7cf	CRITICALITY Non-Critical	VERIFY Test
PARENTS RQTK-SYS-0004 · RQTK-SYS-0008			

VERIFICATION ACTIVITIES

VA-CLI-004-01 Coverage gap report test

Planned

Expected: Requirement ID present in output when gap exists; no IDs in output when no gaps exist.

RQTK-CLI-0005 Graph Export

Approved

Medium

Functional

STATEMENT

The rqt graph command shall export the traceability graph in Graphviz DOT format and write the output to stdout unless a file path is provided via --output.

RATIONALE

DOT is the de-facto interchange format for directed graphs; it can be rendered by Graphviz, Mermaid converters, and CI visualisation pipelines without additional tooling installed alongside rqt.

CATEGORY CLI	HASH 02f35acb	CRITICALITY Non-Critical	VERIFY Test
PARENTS RQTK-SYS-0003			

VERIFICATION ACTIVITIES

VA-CLI-005-01 DOT format validation

Planned

Expected: Valid DOT; expected edge present.

RQTK-CLI-0006 Requirement Export Serialization

Approved

High

Functional

STATEMENT

The rqt export command shall serialize the validated requirement set to the format selected by --format, supporting JSON, CSV, and Markdown.

RATIONALE

Exporting only the validated set prevents corrupted or policy-violating data from propagating into downstream tools and documents.

CATEGORY CLI	HASH 672fa557	CRITICALITY Non-Critical	VERIFY Test
PARENTS RQTK-SYS-0005			

VERIFICATION ACTIVITIES**VA-CLI-006-01** Export format correctness test

Planned

Expected: All three formats valid; each contains the correct number of requirement entries.

RQTK-CLI-0007 Baseline Version Validation

Approved

Medium

Functional

STATEMENT

The rqt baseline command shall exit non-zero without modifying any project files when the provided version string does not conform to the Semantic Versioning 2.0.0 specification.

RATIONALE

Baselines are formal configuration-management milestones; rejecting non-semver strings prevents freeform version labels that break sorting and automated version comparisons from entering the project record.

CATEGORY CLI	HASH b4f2fe32	CRITICALITY Non-Critical	VERIFY Test
PARENTS RQTK-SYS-0001			

VERIFICATION ACTIVITIES**VA-CLI-007-01** Invalid semver rejection test

Planned

Expected: Non-zero exit code; rqt.toml SHA-256 unchanged.

RQTK-CLI-0008 Version Diff

Approved

Medium

Functional

STATEMENT

The rqt diff command shall, given two semver version strings previously recorded by rqt baseline, report all requirements whose history array contains an entry for one version but not the other, indicating additions and removals between those baselines.

RATIONALE

Change-control boards need a concise delta between baselines to assess scope; a diff command removes the need to manually compare two snapshots of a large requirement set.

CATEGORY CLI	HASH c4840c50	CRITICALITY Non-Critical	VERIFY Test
PARENTS RQTK-SYS-0001			

VERIFICATION ACTIVITIES**VA-CLI-008-01** Diff addition detection test

Planned

Expected: R listed under additions; removals section empty.

VA-CLI-008-02 Diff removal detection test

Planned

Expected: S listed under removals; additions section empty.

RQTK-CLI-0009 Lint Command Exit Codes

Approved

Critical

Interface

STATEMENT

The rqt lint command shall exit with code 0 when no violations are present and with code 2 when at least one Error-severity violation is present.

RATIONALE

A predictable exit code allows CI pipelines to gate merges on requirement quality; code 2 (not 1) is chosen so that standard shell errors (exit 1) remain distinguishable from lint failures.

CATEGORY CLI	HASH 585a476b	CRITICALITY Non-Critical	VERIFY Test
PARENTS RQTK-SYS-0004 · RQTK-SYS-0008			

VERIFICATION ACTIVITIES**VA-CLI-009-01** Exit code test

Planned

Expected: Exit codes 0 and 2 respectively.

RQTK-CLI-0010 Coverage Command Exit Code

Approved

High

Interface

STATEMENT

The rqt coverage command shall exit with code 3 when at least one coverage gap is present and with code 0 when all requirements have a non-empty success_criteria field and at least one verification activity.

RATIONALE

Exit code 3 is distinct from lint's exit code 2 and the shell error code 1, allowing CI pipelines to distinguish coverage failures from lint failures and process errors independently.

CATEGORY	HASH	CRITICALITY	VERIFY
CLI	3970d6b0	Non-Critical	Test
PARENTS			
RQTK-SYS-0004			

VERIFICATION ACTIVITIES

VA-CLI-010-01 Coverage exit code test

Planned

Expected: Exit code 3 with gap present; exit code 0 without.

RQTK-CLI-0011 Export Output Destination

Approved

High

Interface

STATEMENT

The rqt export command shall write serialized output to stdout when no output path is specified and to the path provided by --output when specified.

RATIONALE

Defaulting to stdout keeps the command composable with shell pipelines; --output enables writing directly to a file for use in CI artefact collection without shell redirection.

CATEGORY	HASH	CRITICALITY	VERIFY
CLI	396b8390	Non-Critical	Test
PARENTS			
RQTK-SYS-0005			

VERIFICATION ACTIVITIES

VA-CLI-011-01 Export output destination test

Planned

Expected: Stdout contains JSON without --output; file contains JSON and stdout is empty with --output.

RQTK-CLI-0012 Baseline Record Update

Approved

Medium

Functional

STATEMENT

The rqt baseline command shall, upon receiving a version string that conforms to the Semantic Versioning 2.0.0 specification, update the project.version field in rqt.toml to that version and record the baseline timestamp in the same file.

RATIONALE

Updating the project config and recording the timestamp in a single atomic step prevents a partially-applied baseline (version updated but no timestamp) from entering the project record.

CATEGORY	HASH	CRITICALITY	VERIFY
CLI	f795154a	Non-Critical	Test
PARENTS			
RQTK-SYS-0001			

VERIFICATION ACTIVITIES

VA-CLI-012-01	Baseline update test	Planned
Expected: project.version field equals '2.0.0'; baseline timestamp is a valid ISO 8601 datetime.		

4 CORE — Core Library

Requirements for the rqt-core library crate

RQTK-CORE-0001 Validation-Gated API

ApprovedHighDesign

STATEMENT

rqt-core shall prevent coverage queries and graph exports on requirement sets that have not successfully completed the validation pass.

RATIONALE

A guard on operation order eliminates an entire class of misuse: querying coverage or exporting a graph before validation would produce results that silently ignore policy violations. Enforcing this at the API level means the misuse cannot occur without deliberate circumvention.

CATEGORY
CORE

HASH
5aad8787

CRITICALITY
Non-Critical

VERIFY
Analysis

PARENTS
RQTK-SYS-0004

VERIFICATION ACTIVITIES

VA-CORE-001-01 API enforcement analysis

Planned

Expected: Methods absent on unvalidated type; program is rejected (compile-time or runtime error depending on enforcement mechanism).

RQTK-CORE-0002 Graph Cycle Detection

ApprovedCriticalFunctional

STATEMENT

rqt-core shall emit a lint issue with code RQ017 for each cycle detected in the parent-child traceability graph.

RATIONALE

A cyclic dependency between requirements is logically incoherent (A derives from B which derives from A) and would cause infinite loops in any traversal that assumes a DAG; early detection prevents downstream tool failures.

CATEGORY
CORE**HASH**
5d688cad**CRITICALITY**
Non-Critical**VERIFY**
Test**PARENTS**
[RQTK-SYS-0007](#)**VERIFICATION ACTIVITIES****VA-CORE-002-01** Cycle detection unit test

Planned

Expected: LintIssue { code: RQ017, severity: Error } returned.

RQTK-CORE-0003 Auto-Increment ID Generation

Approved

High

Functional

STATEMENT

rqt-core shall generate the next available requirement ID for a given category by scanning the loaded requirement set for the highest existing ordinal within that category and formatting the result with the zero-padding width specified in the project configuration.

RATIONALE

Centralising ID generation in the core library ensures the CLI and any future integrations (IDE plugins, web UI) produce IDs that are consistent with the project's configured padding and prefix.

CATEGORY
CORE**HASH**
e82dfec**CRITICALITY**
Non-Critical**VERIFY**
Test**PARENTS**
[RQTK-SYS-0001](#) · [RQTK-SYS-0002](#)**VERIFICATION ACTIVITIES****VA-CORE-003-01** ID generation unit test

Planned

Expected: Generated ID equals 'RQTK-SYS-0003'.

RQTK-CORE-0004 Config-Driven Policy Engine

Approved

High

Design

STATEMENT

rqt-core shall derive all validation rules — including allowed categories, types, states, priorities, verification methods, and parent-hierarchy constraints — exclusively from the ProjectConfig provided at library initialization.

RATIONALE

Config-driven policy lets any project (aerospace, automotive, software-only) enforce its own vocabulary without forking the library. The corollary is that the library ships with no domain vocabulary of its own; a project that omits a field from its config receives no default policy for that field.

CATEGORY	HASH	CRITICALITY	VERIFY
CORE	06404eb5	Non-Critical	Test
PARENTS			
RQTK-SYS-0004			

VERIFICATION ACTIVITIES

VA-CORE-004-01	Custom category acceptance test	Planned
Expected: Exit 0 with config; exit 2 with RQ002 without config.		

5 Traceability Matrix

ID	Title	Category	Parent(s)
RQTK-CLI-0001	Add Requirement	CLI	RQTK-SYS-0001, RQTK-SYS-0002
RQTK-CLI-0002	Lint Command Output	CLI	RQTK-SYS-0004, RQTK-SYS-0008
RQTK-CLI-0003	Traceability Query	CLI	RQTK-SYS-0003
RQTK-CLI-0004	Coverage Gap Detection	CLI	RQTK-SYS-0004, RQTK-SYS-0008
RQTK-CLI-0005	Graph Export	CLI	RQTK-SYS-0003
RQTK-CLI-0006	Requirement Export Se- rialization	CLI	RQTK-SYS-0005
RQTK-CLI-0007	Baseline Version Valida- tion	CLI	RQTK-SYS-0001
RQTK-CLI-0008	Version Diff	CLI	RQTK-SYS-0001
RQTK-CLI-0009	Lint Command Exit Codes	CLI	RQTK-SYS-0004, RQTK-SYS-0008
RQTK-CLI-0010	Coverage Command Exit Code	CLI	RQTK-SYS-0004
RQTK-CLI-0011	Export Output Destina- tion	CLI	RQTK-SYS-0005
RQTK-CLI-0012	Baseline Record Update	CLI	RQTK-SYS-0001
RQTK-CORE-0001	Validation-Gated API	CORE	RQTK-SYS-0004
RQTK-CORE-0002	Graph Cycle Detection	CORE	RQTK-SYS-0007
RQTK-CORE-0003	Auto-Increment ID Generation	CORE	RQTK-SYS-0001, RQTK-SYS-0002
RQTK-CORE-0004	Config-Driven Policy Engine	CORE	RQTK-SYS-0004
RQTK-SYS-0001	Requirement Lifecycle Management	SYS	—
RQTK-SYS-0002	TOML-Based File Stor- age	SYS	—
RQTK-SYS-0003	Requirement Traceabil- ity Graph	SYS	—
RQTK-SYS-0004	Policy-Driven Valida- tion	SYS	—
RQTK-SYS-0005	Multi-Format Export	SYS	—

RQTK-SYS-0006	VCS Merge Compatibility	SYS	—
RQTK-SYS-0007	Traceability Graph Defect Detection	SYS	—
RQTK-SYS-0008	Structured Violation Reporting	SYS	—

6 Verification Summary

ID	Title	Method	Level	Activities
RQTK-CLI-0001	Add Requirement	Test	Integration	1
RQTK-CLI-0002	Lint Command Output	Test	Integration	1
RQTK-CLI-0003	Traceability Query	Test	Integration	1
RQTK-CLI-0004	Coverage Gap Detection	Test	Integration	1
RQTK-CLI-0005	Graph Export	Test	Integration	1
RQTK-CLI-0006	Requirement Export Serial- ization	Test	Integration	1
RQTK-CLI-0007	Baseline Version Valida- tion	Test	Integration	1
RQTK-CLI-0008	Version Diff	Test	Integration	2
RQTK-CLI-0009	Lint Command Exit Codes	Test	Integration	1
RQTK-CLI-0010	Coverage Command Exit Code	Test	Integration	1
RQTK-CLI-0011	Export Output Destina- tion	Test	Integration	1
RQTK-CLI-0012	Baseline Record Update	Test	Integration	1
RQTK-CORE-0001	Validation-Gated API	Analysis	Unit	1
RQTK-CORE-0002	Graph Cycle Detection	Test	Unit	1
RQTK-CORE-0003	Auto-Increment ID Gen- eration	Test	Unit	1
RQTK-CORE-0004	Config-Driven Policy En- gine	Test	Integration	1
RQTK-SYS-0001	Requirement Lifecycle Management	Test	System	2
RQTK-SYS-0002	TOML-Based File Storage	Inspection	System	1
RQTK-SYS-0003	Requirement Traceability Graph	Test	System	1
RQTK-SYS-0004	Policy-Driven Validation	Test	System	1
RQTK-SYS-0005	Multi-Format Export	Test	System	1
RQTK-SYS-0006	VCS Merge Compatibility	Demonstration	System	1

RQTK-SYS-0007	Traceability Graph Defect Detection	Test	System	3
RQTK-SYS-0008	Structured Violation Reporting	Test	System	1